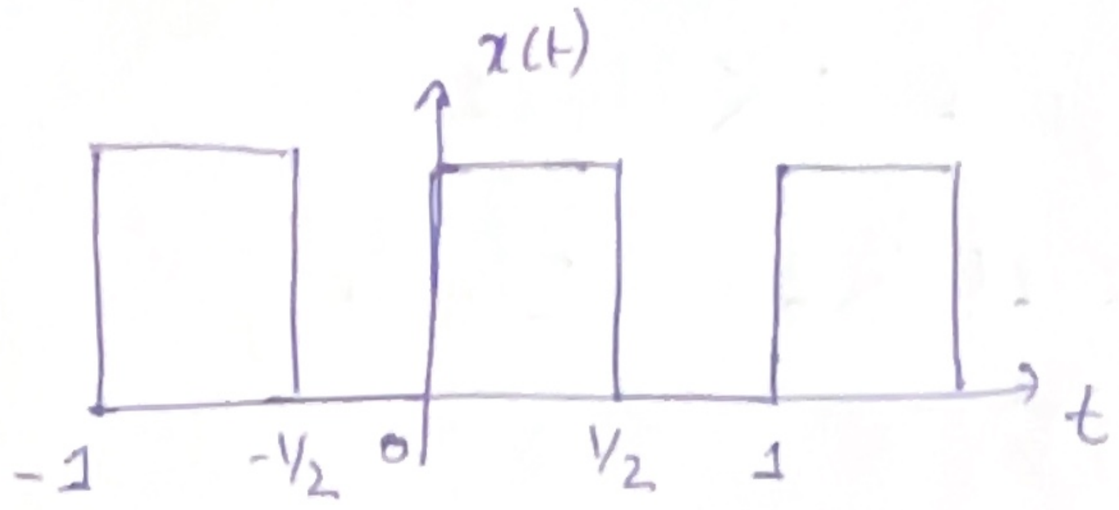
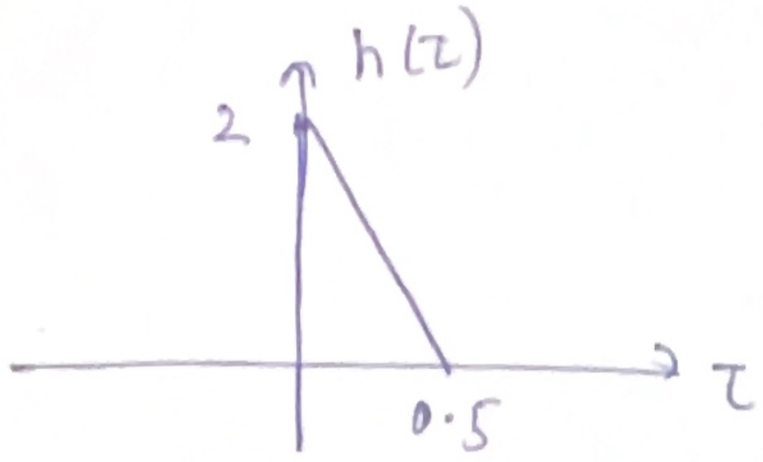
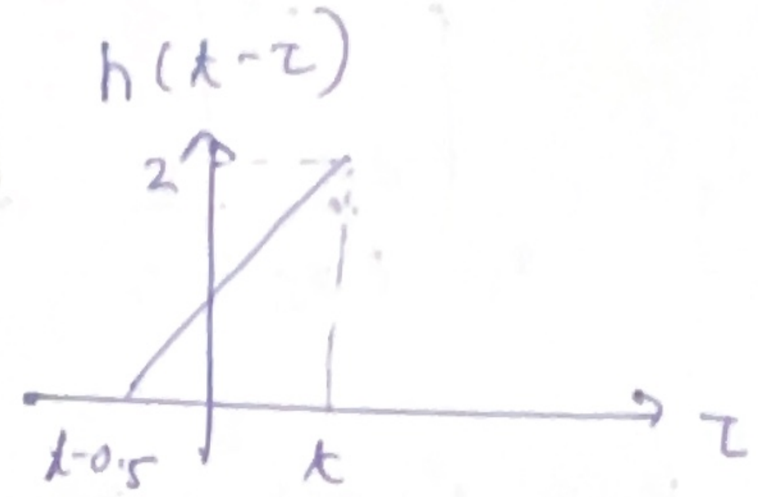
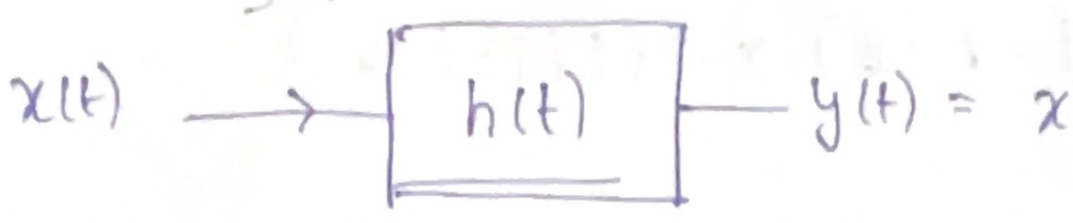
Q.1

a)









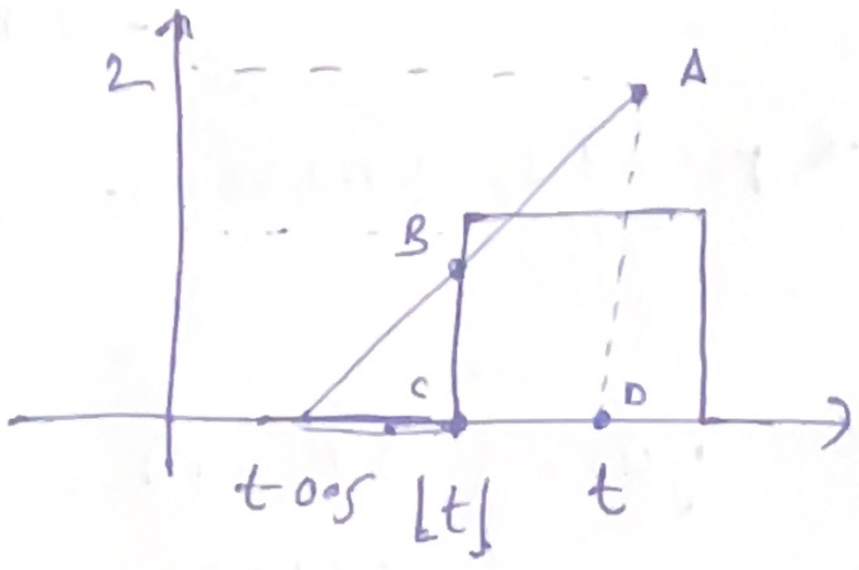
1 marks

Let $\lfloor x \rfloor$ denote greatest integer $\leq x$ & $\{x\} = x - \lfloor x \rfloor$ fractional part of x

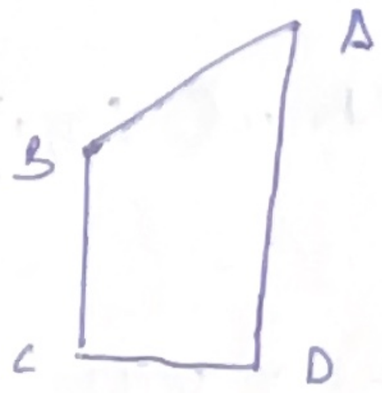
Consider the range $t - \frac{1}{2} \leq \lfloor t \rfloor < t$

$\Rightarrow 0 \leq \{t\} \leq 0.5$

1 marks



$y(t)$ = Area of trapezium ABCD



$AD = 2$

$CD = t - \lfloor t \rfloor = \{t\}$

$BC = 4(\lfloor t \rfloor - t + 0.5)$

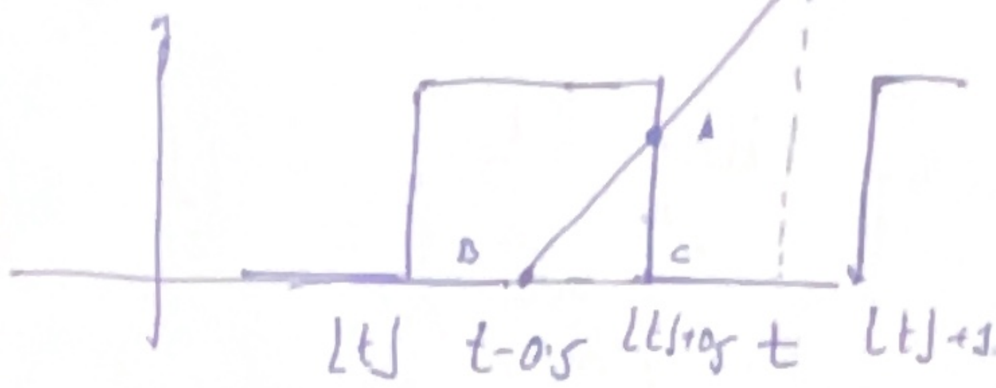
$\Rightarrow y(t) = \frac{1}{2}(4(0.5 - \{t\}) + 2)\{t\}$

2 marks

$= 2\{t\}(1 - \{t\}), \quad 0 \leq \{t\} \leq 0.5$

Now consider $t \geq \lfloor t \rfloor + 0.5$ & $t \leq \lfloor t \rfloor + 1$.

$\Rightarrow \quad 0.5 \leq \{t\} < 1$. - 1 mark



$y(t)$ = area of ΔABC

$= \frac{1}{2} \times (\lfloor t \rfloor - t + 1) \times 1 (\lfloor t \rfloor - t + 1)$

$= 2 (1 - \{t\})^2$, $0.5 \leq \{t\} < 1$. - 2 marks

$$y(t) = \begin{cases} 2\{t\}(1-\{t\}), & 0 \leq \{t\} < 0.5 \\ 2(1-\{t\})^2, & 0.5 \leq \{t\} < 1. \end{cases}$$

- 1 marks

Since $\{t\}$ is periodic with period 1, $y(t)$ is also periodic with period.

In the interval $[0,1)$ $\quad y(t) = \begin{cases} 2t(1-t), & 0 \leq t < 0.5 \\ 2(1-t)^2, & 0.5 \leq t < 1 \end{cases}$

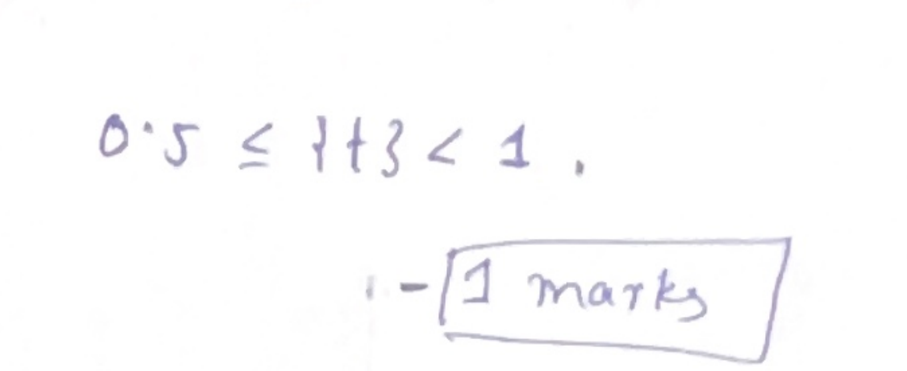
Max value of $y(t) = 0.5$ - 1 marks

Min value of $y(t) = 0$ - 1 marks

Time period = 1 - 0.5 marks

Angular frequency = $\frac{2\pi}{T} = 2\pi$.

- 0.5 marks



- 1 marks

Q1 b)

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \qquad \omega_0 = 2\pi T \Rightarrow 2\pi$$

$$= \sum_{k=-\infty}^{\infty} a_k e^{jk2\pi t}$$

$$\Rightarrow a_k = \frac{1}{T}\int_0^T x(t)\, e^{-jk\omega_0 t}\, dt$$

$$= 1 \int_0^1 x(t)\, e^{-jk2\pi t}\, dt$$

$$= \int_0^{1/2} 1 \cdot e^{-jk2\pi t}\, dt + \int_{1/2}^1 0 \cdot e^{-jk2\pi t}\, dt$$

$$\Rightarrow a_k = \int_0^{1/2} e^{-jk2\pi t}\, dt$$

case 0 $\Rightarrow$ $k = 0$

$$a_0 = \int_0^{1/2} 1\, dt \Rightarrow a_0 = 1/2 \quad \text{—— } 1/4$$

~~Case~~ $k \neq 0$

$$a_k = \int_0^{1/2} e^{-j2\pi kt}\, dt$$

$$= \frac{\left[e^{-jk\pi} - 1\right]}{-j2\pi k} \qquad \left[e^{-jk\pi} = (-1)^k\right]$$

$$= \frac{(-1)^k - 1}{-j2\pi k}$$

k = even $\Rightarrow (-1)^k = 1$

$\Rightarrow a_k = 0$

k = odd $\Rightarrow (-1)^k = -1$

$$\Rightarrow a_k = \frac{-2}{-j2\pi k}$$

$$= \frac{1}{j\pi k} \quad \text{—— } 1$$

[Total 1·25 for a_k]

$$\Rightarrow a_k = \begin{cases} 1/2 & k=0 \quad -0.25 \\ 0 & k=2i \quad i \in \mathbb{Z} \quad -0.5 \\ \frac{1}{j\pi k} & k=2i+1 \quad i \in \mathbb{Z} \quad -0.5 \end{cases}$$

a_k is the kth F.S coeff.

Standard Representation

$$x(t) = \frac{a_0}{2} + \sum_{k=1}^{\infty} a_k \cos(k\omega_0 t) + b_k \sin(k\omega_0 t)$$

$$= \sum_{k=1}^{\infty} a_k e^{jk\omega_0 t} + a_{-k} e^{-jk\omega_0 t} + a_0$$

Standard Representation

$$x(t) = \frac{a_0}{2} + \sum_{k=1}^{\infty} a_k \cos(k\omega_0 t) + b_k \sin(k\omega_0 t)$$

$$= \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t}$$

$$\frac{a_0}{2} = c_0 = \frac{1}{2} \Rightarrow a_0 = 1 \quad -0.25$$

$$c_k = \frac{a_k - j b_k}{2} \quad \text{and} \quad c_{-k} = \frac{a_k + j b_k}{2} \quad [k>0]$$

$$\Rightarrow a_k = c_k + c_{-k} \quad ; \quad b_k = j(c_k - c_{-k}) \quad -1$$

[1 mark for correct relation]

$c_k = 0$ for ~~even~~ even k

$\Rightarrow a_k = b_k = 0$ $\quad k$ = even. — 0.5

$c_k = \frac{1}{2j\pi k}$ for odd k

$\Rightarrow a_k = \frac{1}{j\pi k} + \frac{1}{j\pi(-k)} = 0$ — 0.25

$b_k = j\left[\frac{1}{j\pi k} - \frac{1}{j\pi(-k)}\right]$

$= \frac{2}{\pi k}$ — 0.25

$\Rightarrow x(t) = \frac{1}{2} + \frac{2}{\pi}\left[\sin(2\pi t) + \frac{1}{3}\sin(6\pi t) + \frac{1}{5}\sin(10\pi t) \cdots\right]$

[Total 1.25]

$= \frac{1}{2} + \sum_{\substack{k=1 \\ \text{odd}}}^{\infty} \frac{2}{\pi k}\sin(2\pi k t)$

$= \frac{1}{2} + \sum_{n=0}^{\infty} \frac{2}{\pi(2n+1)}\sin(2\pi(2n+1)t)$

Fourier Series of $y(t)$

$y(t) = \sum Y_k e^{jk\omega_0 t}$

$x(t) = \sum X_k e^{jk\omega_0 t}$

$Y_k = H(jk\omega_0) X_k$ — 1

$H(0) = \frac{1}{2}$ $\quad H(jk\omega_0) = \frac{-j}{\pi k} + \frac{1-(-1)^k}{\pi^2 k^2}$

$\Rightarrow Y_0 = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$

$k \neq 0$

$$Y_k = \left(\frac{-j}{\pi k} + \frac{1-(-1)^k}{\pi^2 k^2} \right) \left(\frac{1-(-1)^k}{j 2\pi k} \right) \quad -1$$

$$Y_k = \begin{cases} 1/4 & k=0 \quad -0.25 \\ 0 & \text{even } k \quad -0.5 \\ \frac{-1}{\pi^2 k^2} - \frac{2j}{\pi^3 k^3} & \text{odd } k \quad -0.5 \end{cases}$$

[Total 1.25 for Y_k]

[Total 1.25]

Standard Trignometric Rrepresentation

$$y(t) = a_0/2 + \sum a_k \cos(2\pi k t) + b_k \sin(2\pi k t) \qquad [\omega_0 = 1]$$

$$Y_k = \frac{a_k - j b_k}{2} \quad \forall \quad \frac{a_0}{2} = Y_0 = \frac{1}{4} \quad -0.25$$

$k = \text{even} \Rightarrow Y_k = 0 \Rightarrow a_k = b_k = 0 \quad -0.5$

$$k = \text{odd} \Rightarrow \frac{a_k - j b_k}{2} = \frac{-1}{\pi^2 k^2} - \frac{2j}{\pi^3 k^3}$$

$$\Rightarrow a_k = \frac{-2}{\pi^2 k^2}, \quad b_k = \frac{4}{\pi^3 k^3} \quad -0.25 \quad -0.25$$

$$\Rightarrow y(t) = \frac{1}{4} + \sum_{\substack{k=1 \\ \text{odd}}}^{\infty} \left[\frac{-2}{\pi^2 k^2} \cos 2\pi k t + \frac{4}{\pi^3 k^3} \sin 2\pi k t \right]$$

[Total 1.25]

[Any other method is valid as long as mathematically correct, some results are directly used which are commonly known]

① (c) **Def^n**: A system is Causal if the o/p at any time t_0 depends only on i/p at times $t \le t_0$.

For an LTI it is equivalent to:

$$h(t) = 0 \quad \forall\, t < 0$$

From the graph, $h(t) = 0 \quad \forall\, t < 0$ ⟶ Hence, Causal.

Def^n: A system is BIBO stable if every bounded i/p produces bounded o/p.

For an LTI, $\int_{-\infty}^{\infty} |h(t)|\, dt < \infty$

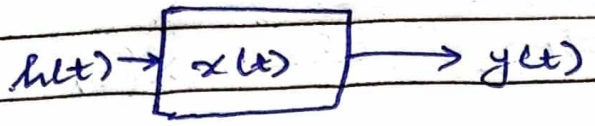
Clearly, $\int_{-\infty}^{\infty} |h(t)|\, dt$ ⟶ finite from graph

⟶ Hence, BIBO stable.

Def^n ⟶ 0.5 each x2 = 1M

justification ⟶ 1.5 each x 2 = 3M

Q.1 (d) ⇒



⇒ $y(t) = h(t) * x(t)$

& by commutative property

⇒ $h(t) * x(t) = x(t) * h(t)$ —— (1) mark

∴ $y(t) = x(t) * h(t)$

& from part 1 (d)

⇒ $y(t) = x(t) * h(t)$

$$= \begin{cases} 2\{t\}(1-\{t\}) ; & 0 \le \{t\} < 0.5 \\ 2(1-\{t\})^2 ; & 0.5 \le \{t\} < 1 \end{cases}$$

—— (1) mark

Causality

⇒ $x(t)$ → periodic func or periodic pulse train

&

$x(t) \neq 0$ for $x < 0$

∴ The system is non-causal. —— (1) mark

Stability

⇒ BIBO stability condⁿ :

$$\int_{-\infty}^{\infty} |x(t)|\, dt < \infty$$

& $\int_{0}^{1} |x(t)|\, dt = 0.5$ for a period

& t extends from $-\infty$ to ∞ for periodic signal $x(t)$.

∴ Adding this result for infinite no. of intervals (whole range of t) gives ∞.

⇒ ∴ $\int_{-\infty}^{\infty} |x(t)|\, dt \to \infty$

∴ The system is unstable. —— (1) mark

⇨ As O/P (y(t)) for both the LTI system is same because of commutative property but there impulse response are different & one system is causal & stable while other is neither caused nor stable. Hence, we can't say anything about the internal properties (causal, stable etc.) of the system, just by observing its o/p for one particular I/P.

—— (1) mark

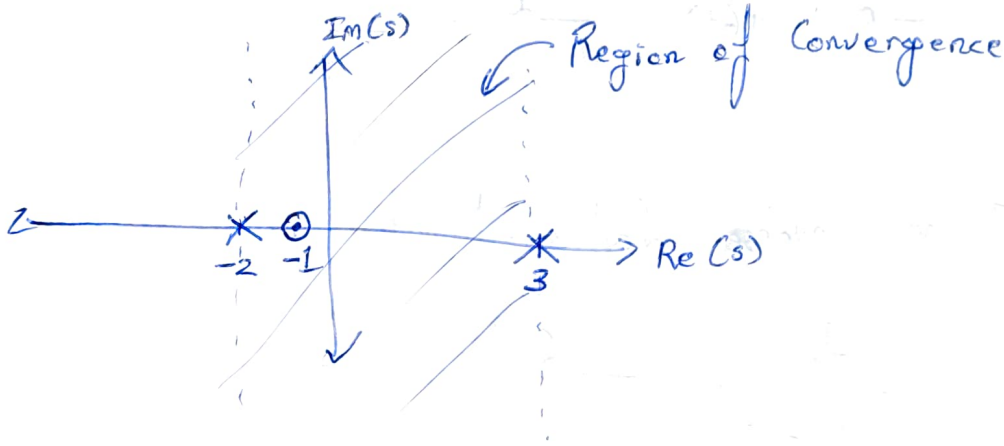
QUESTION 2

a) $$X(s) = \frac{s+1}{s^2 - s - 6} = \frac{s+1}{s^2 - 3s + 2s - 6}$$

$$x(s) = \frac{(s+1)}{s(s-3) + 2(s-3)} = \frac{(s+1)}{(s+2)(s-3)}$$

Zeros : -1

Poles : $-2, +3$



$$X(s) = \frac{A}{s+2} + \frac{B}{s-3} = \frac{(s+1)}{(s+2)(s-3)}$$

$$\frac{A(s-3) + B(s+2)}{(s+2)(s-3)} = \frac{(s+1)}{(s+2)(s-3)}$$

$$A(s-3) + B(s+2) = (s+1)$$

PUTTING $s = 3$ $\quad B(s+2) = (s+1)$

$B(5) = 4$

$$\boxed{B = 0.8 \text{ or } 4/5}$$

PUTTING $s = -2$ $\quad A(-5) = -1$

$$\boxed{A = 0.2 = 1/5}$$

$$X(s) = \frac{1/5}{(s+2)} + \frac{4/5}{(s-3)} \quad \text{ROC} : -2 < \text{Re}\{s\} < 3$$

Using,

$$e^{-\alpha t} u(t) \xrightarrow{\mathcal{L}} \frac{1}{s+\alpha} \quad \text{for } \text{Re}(s) > -\alpha$$

(and)

$$-e^{-\alpha t} u(-t) \xrightarrow{\mathcal{L}} \frac{1}{s+\alpha} \quad \text{for } \text{Re}(s) < -\alpha$$

Taking inverse Laplace transform of $X(s)$

$$\frac{\frac{1}{5}}{(s+2)} \xrightarrow{\mathcal{L}^{-1}} \frac{1}{5} e^{-2t} u(t) \quad \because \text{Re}(s) > -2$$

and,

$$\frac{\frac{4}{5}}{s-3} \xrightarrow{\mathcal{L}^{-1}} -\frac{4}{5} e^{3t} u(-t) \quad \because \text{Re}(s) < 3$$

Thus,

$$x(t) = \frac{1}{5} e^{-2t} u(t) - \frac{4}{5} e^{3t} u(-t)$$

Marking Scheme :

- 1M for finding zero
- 1M each for finding poles
- 1.5M for final answer of $x(t)$
- 1.5M for showing any valid approaches

b) Yes, $x(t)$ has a valid FT

Expl: Since the $s = j\omega$ axis lies in the ROC, it satisfies the condition for the existence of FT

to find the FT substitute $s = j\omega$ in the $X(s)$ expression

$$X(s) = \frac{(s+1)}{(s+2)(s-3)}$$

$$X(j\omega) = \frac{(1+j\omega)}{-\omega^2 - j\omega - 6} = -\frac{(1+j\omega)}{\omega^2 + j\omega + 6}$$

- Marking Scheme:
- 1M for YES
- 1M for proper explanation.
- 2M to substitute $s = j\omega$ and simplifying

Question 3(a) — Sanjana

1st part [4M]

Given,

$$y(t) = \frac{d(x(t))}{dt} \longrightarrow \text{Differentiator}$$

Linearity:

let $x_1(t) \longrightarrow y_1(t)$

$x_2(t) \longrightarrow y_2(t)$

Now, let $x_3(t) = \alpha x_1(t) + \beta x_2(t)$

$x_3(t) \longrightarrow y_3(t)$

$$y_3(t) = \frac{d}{dt}[x_3(t)]$$

$$= \frac{d}{dt}[\alpha x_1(t) + \beta x_2(t)]$$

$$= \alpha \frac{d}{dt}(x_1(t)) + \beta \frac{d}{dt}(x_2(t))$$

$$= \alpha y_1(t) + \beta y_2(t)$$

$\therefore y_3(t) = \alpha y_1(t) + \beta y_2(t) \longrightarrow$ System is linear [2M]

Time-Invariance

→ let for arbitary value of t_0,

$$x(t-t_0) \longrightarrow y_1(t)$$

A system is time-invariant if

$$y_1(t) = y(t-t_0)$$

So,

→ $y_1(t) = \frac{d}{dt}[x(t-t_0)]$

→ $y(t-t_0) = \frac{d}{dt}[x(t-t_0)]$

$y_1(t) = y(t-t_0)$

$\therefore$ System is time-invariant [2M]

$\therefore$ A differentiator is LTI system

2nd Part [4M]

Given,

$$x(t) = \frac{1}{2\pi j}\int_{\omega-j\infty}^{\omega+j\infty} X(s)e^{st}\,ds$$

→ Differentiate on both sides wrt t,

$$\frac{d}{dt}[x(t)] = \frac{d}{dt}\left[\frac{1}{2\pi j}\int_{\omega-j\infty}^{\omega+j\infty} X(s)e^{st}\,ds\right] \quad \Big\} [2M]$$

$$= \frac{1}{2\pi j}\int_{\omega-j\infty}^{\omega+j\infty} X(s)\frac{\partial}{\partial t}(e^{st})\,ds \longrightarrow \frac{\partial}{\partial t}(e^{st}) = se^{st}$$

$$= \frac{1}{2\pi j}\int_{\omega-j\infty}^{\omega+j\infty} sX(s)e^{st}\,ds$$

But $y(t) = \frac{d}{dt}[x(t)]$

$$\therefore\ y(t) = \frac{1}{2\pi j}\int_{\omega-j\infty}^{\omega+j\infty} \underbrace{[sX(s)]}_{Y(s)}e^{st}\,ds \longrightarrow [1M]$$

By uniqueness of inverse laplace transform,

$$Y(s) = sX(s) \longrightarrow [1M]$$

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Question 3(b)

Q. $n \in \mathbb{Z}^+$. There are n differentiators connected in series. If $Y(s) = \frac{s^{n-1}}{(s+2)}$ is the Laplace transform of the output, and Laplace transform of the input has $\text{ROC} = \{s : \text{Re}(s) > 0\}$, then find the i/p signal.

A: We know that, by the differentiation property of the Laplace transformation,

if $y(t) = \frac{d}{dt} x(t)$, then $Y(s) = s \cdot X(s)$ (also proved in 3(a))

For n differentiators connected in series, the system will be,

$$y(t) = \frac{d^n}{dt^n} x(t) \quad \text{(ie, the } n^{\text{th}} \text{ derivative of } x(t))$$

The Laplace transformation of $y(t)$ will be given by,

$$Y(s) = \int_{-\infty}^{\infty} y(t) e^{-st} dt = \int_{-\infty}^{\infty} \frac{d^n}{dt^n} x(t) e^{-st} dt$$

Using integration by parts,

$$Y(s) = e^{-st} \int \frac{d^n}{dt^n} x(t) dt - \int (-s) e^{-st} \int \frac{d^n}{dt^n} x(t) dt$$

$$= \left[e^{-st} \frac{d^{n-1}}{dt^{n-1}} x(t) \right]_{-\infty}^{\infty} + s \int \frac{d^{n-1}}{dt^{n-1}} x(t) e^{-st} dt$$

↳ Assume this is zero.

$$\Rightarrow Y(s) = s \cdot \mathcal{L}\left\{ \frac{d^{n-1}}{dt^{n-1}} x(t) \right\}$$

Continuing this, we will get,

$$Y(s) = s^n X(s)$$

Comparing this with the representation of LTI systems in laplace ($Y(s) = X(s) \cdot H(s)$) we get,

$$H(s) = s^n$$

(1.5, any other valid explanation of system function's derivation will also suffice)

In the given scenario, $Y(s) = \frac{s^{n-1}}{s+2}$

$$Y(s) = X(s) H(s)$$

$$\Rightarrow X(s) = \frac{Y(s)}{H(s)} = \frac{s^{n-1}}{s^n (s+2)} = \frac{1}{s(s+2)}$$

Using Partial fraction to split $X(s)$,

$$X(s) = \frac{1}{s(s+2)} = \frac{\frac{1}{2}(s+2-s)}{s(s+2)}$$

$$= \frac{1}{2}\left(\frac{s+2}{s(s+2)} - \frac{s}{s(s+2)}\right)$$

$$X(s) = \frac{1}{2}\left(\frac{1}{s} - \frac{1}{s+2}\right)$$

Applying Inverse laplace Transform,

$$x(t) = \frac{1}{2}\left(\mathcal{L}^{-1}\{\tfrac{1}{s}\} - \mathcal{L}^{-1}\{\tfrac{1}{s+2}\}\right)$$

Using the standard results and the given ROC,

$$\mathcal{L}^{-1}\{1/s\} = u(t)$$

$$\mathcal{L}^{-1}\{1/(s+2)\} = e^{-2t} u(t)$$

$$\Rightarrow x(t) = \frac{1}{2}\left(1 - e^{-2t}\right) u(t)$$

(2.5)